Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech. C.E. and 3rd sem
Name of the Paper: Mechanics of Solids
Paper Code: TCE-303
Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO1

a. State and explain the principles of statics. Discuss their significance in analyzing concurrent forces in a plane.

b. A system of concurrent forces is in equilibrium. Two forces of 50 N and 80 N act at 30° and 120° respectively. Find the magnitude and direction of the third force.

c. Define parallel forces in a plane and explain the conditions for their equilibrium.

Q2. (2X10=20 Marks) CO2

a. Describe the method of joints for analyzing forces in trusses. Provide a step-by-step procedure with an example.

b. Define stress and strain. Explain the differences between axial stress, shear stress, and biaxial state of stress with examples.

c. A compound bar consists of two segments of different materials, joined in series. Determine the stresses and elongations in each material when subjected to a load of 100 kN.

Q3. (2X10=20 Marks) CO3

a. A bar is subjected to normal stresses of σx=80 MPa and σy=50 MPa with a shear stress of τxy=30 MPa. Determine the principal stresses using equations and verify using Mohr's circle.

b. Explain the concept of principal stresses. How are they different from normal and shear stresses?

c. A simply supported beam of span 4 m carries two-point loads of 10 kN and 15 kN at 1 m1 and 3 m from the left support. Calculate reactions at supports and draw the SFD and BMD.

Q4. (2X10=20 Marks) CO4

a. A cantilever beam of length 4 m is subjected to a concentrated load of 20 kN at its free end. Determine the slope and deflection at the free end using singularity functions.

b. A simply supported beam of length 5 m is subjected to a concentrated load of 15 kN at its mid span. Find the deflection at the center using the double integration method.

c. Derive the expression for the deflection and slope of a beam subjected to a point load at any point using Macaulay's method.

Q5. (2X10=20 Marks) CO5

a. Explain the concept of pure torsion and the assumptions made in the theory of pure torsion.

b. Explain the difference between short and long columns. How can you determine the classification of a column as short or long, and what is the importance of this classification in design?

c. Calculate the torsional rigidity of a hollow shaft with an outer diameter of 200 mm and an inner diameter of 100 mm, if the shear modulus of the material is $G=80\times109$ N/m^2.